COMPUTER VISION · FINAL PROJECT

Improving Small-Object Detection in CutLER via Multi-Scale Pseudo-Label Generation

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PROBLEM FORMULATION

CutLER struggles with small objects

Pre-trained CutLER's small-object AP is 30× lower than its large-object AP.

APs (small)

1.3

COCO smal AP

APm (medium)

6.5

COCO medium-object AP

APl (large)

32.2

COCO large-object AP

THE GAP

30×

APl / APs

WHY THIS MATTERS

Small-object detection is critical for autonomous driving, surveillance, and aerial imagery - domains where the objects of interest are often a small fraction of the frame.

BACKGROUND

How CutLER works

STAGE 1 · Pseudo-label generation

STAGE 2 · Detector training

Image
ImageNet input

→

MaskCut
DINO + Normalized Cuts

→

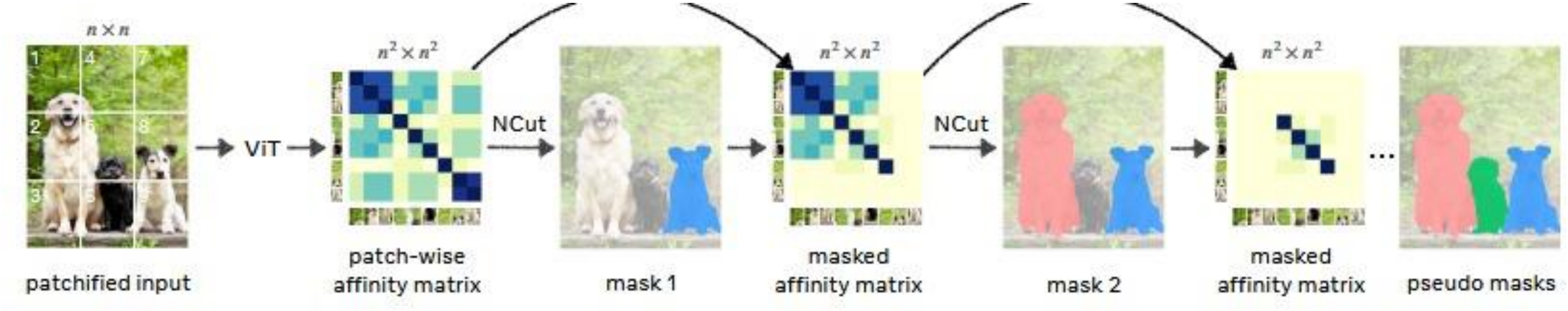
Pseudo-labels
masks per image

→

Cascade Mask R-CNN
trained on pseudo-labels

→

Detector
unsupervised



WHY SINGE-SCALE MASKCUT MISSES SMALL OBJECTS

MaskCut runs on a 14×14 ViT patch grid where small objects occupy only 1–2 patches, leaving DINO with almost no signal to separate them from background. Normalized Cuts then optimizes a global partition that's dominated by large prominent objects, so small ones get absorbed into the background segment and never enter the pseudo-labels.

DATA SOURCING STRATEGY

Training & Evaluation

TRAINING DATA

TinyImageNet subset

10 classes · 500 images · 50 / class

WHY NOT FULL IMAGENET-1K

Retraining the full Cascade Mask R-CNN on 1.3M images requires multi-GPU compute.
A 500-image subset keeps the full pipeline tractable on a single GPU within the course budget

EVALUATION DATA

COCO val2017

Class-agnostic detection benchmark

WHY COCO

Matches the CutLER paper's evaluation protocol, so our results are directly comparable.

COCO's small / medium / large object splits give us a clean view of where our multi-scale pipeline gains or loses ground.

PROPOSED SOLUTION

Multi-scale heatmap-guided MaskCut

Run MaskCut at full image scale, then re-run on heatmap-selected crops where small objects likely hide.

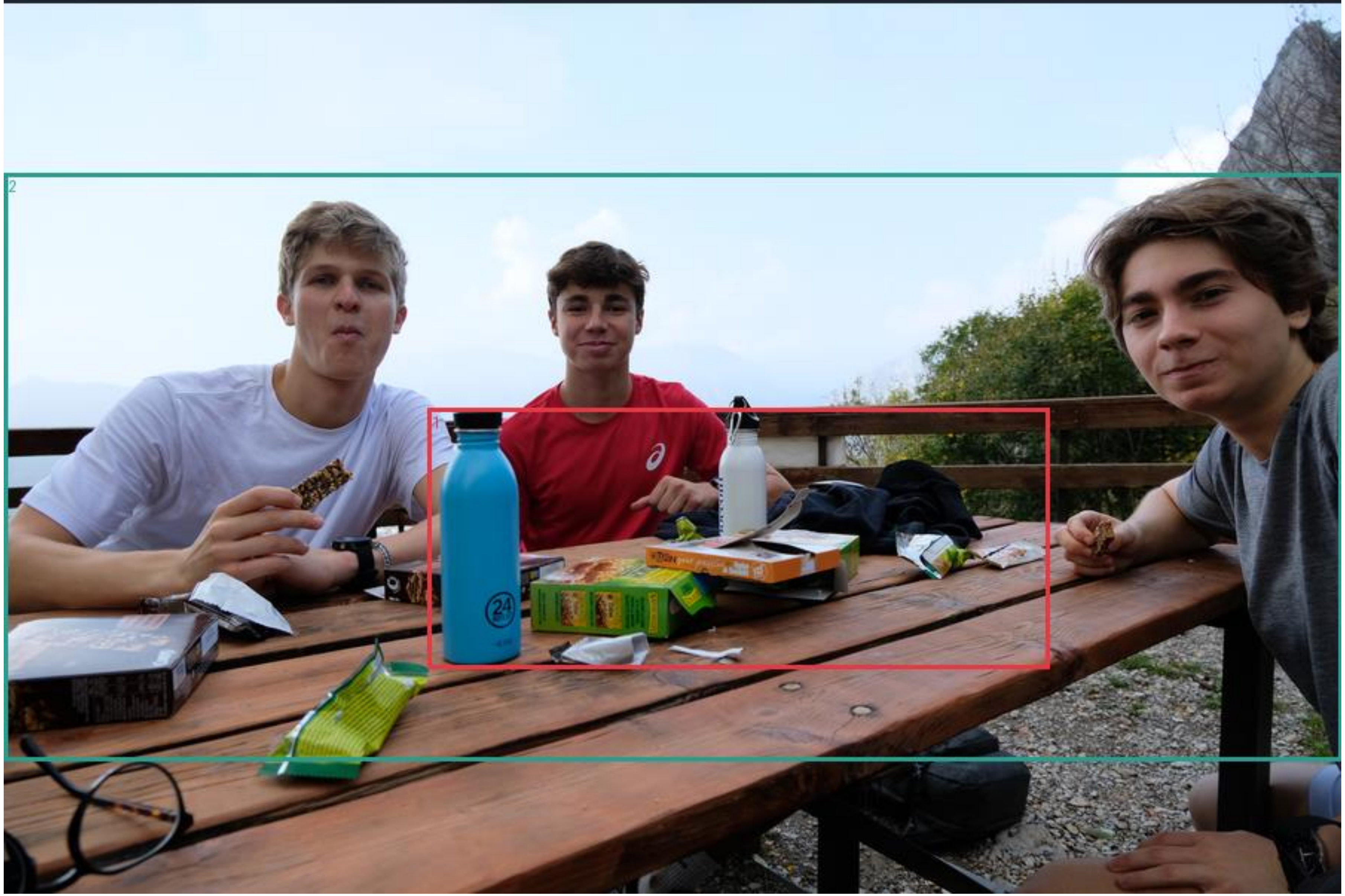
1. **Full-image MaskCut** — baseline pass
2. **DINO heatmap** — feature contrast
3. **Crop selection** — high-score regions
4. **MaskCut per crop** — fine-scale pass
5. **Score & filter** — remove noise and duplicates
6. **Merge** — final pseudo-labels

MECHANISM

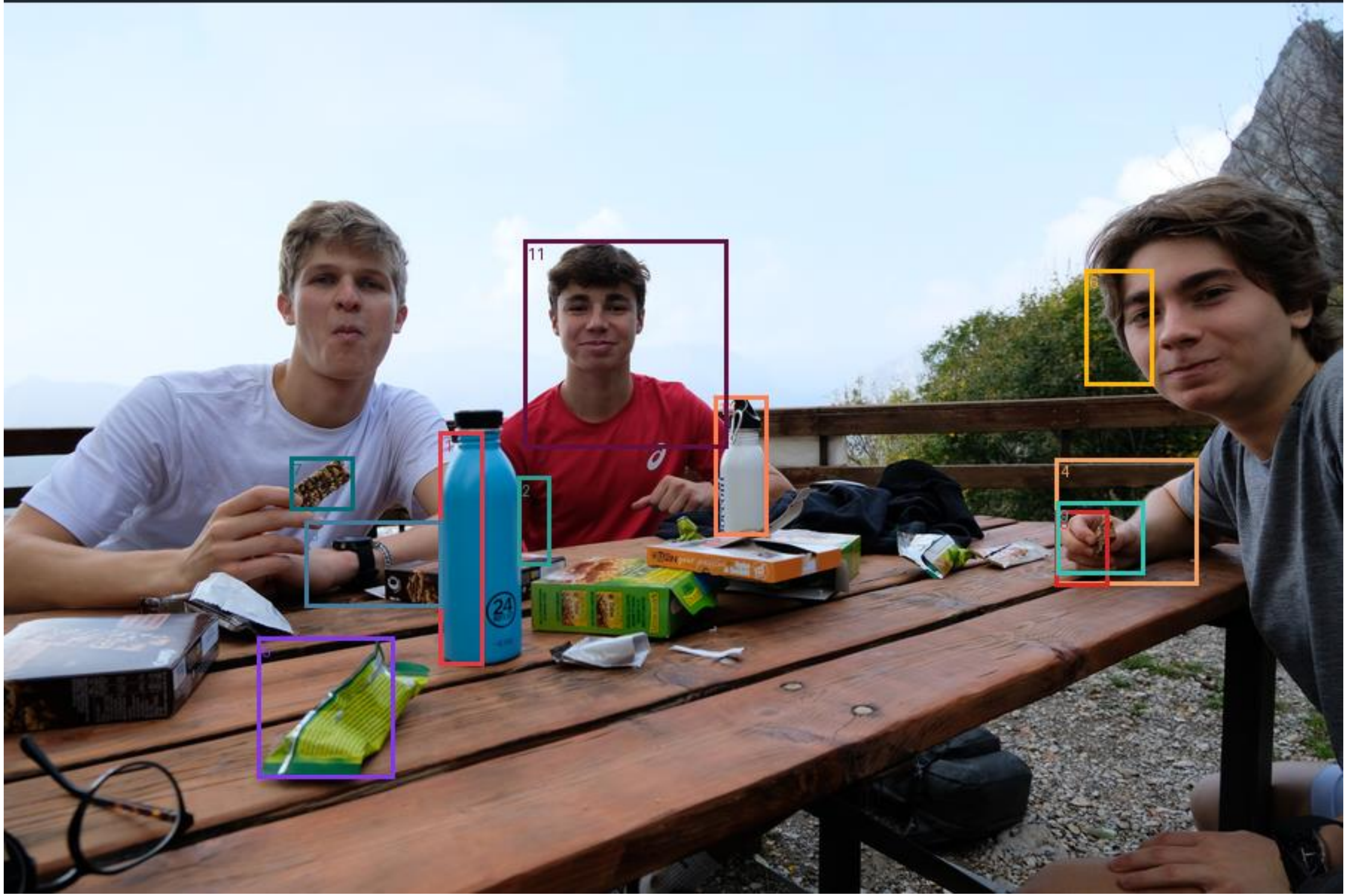
Crops give small objects more ViT patches.

Inside a 1/4-image crop a small object now occupies ~4× more patches.
DINO features become more locally discriminative, so MaskCut can seperate foreground regions that were too small at full image.

normal_full | DSCF0136.JPG | 2 masks

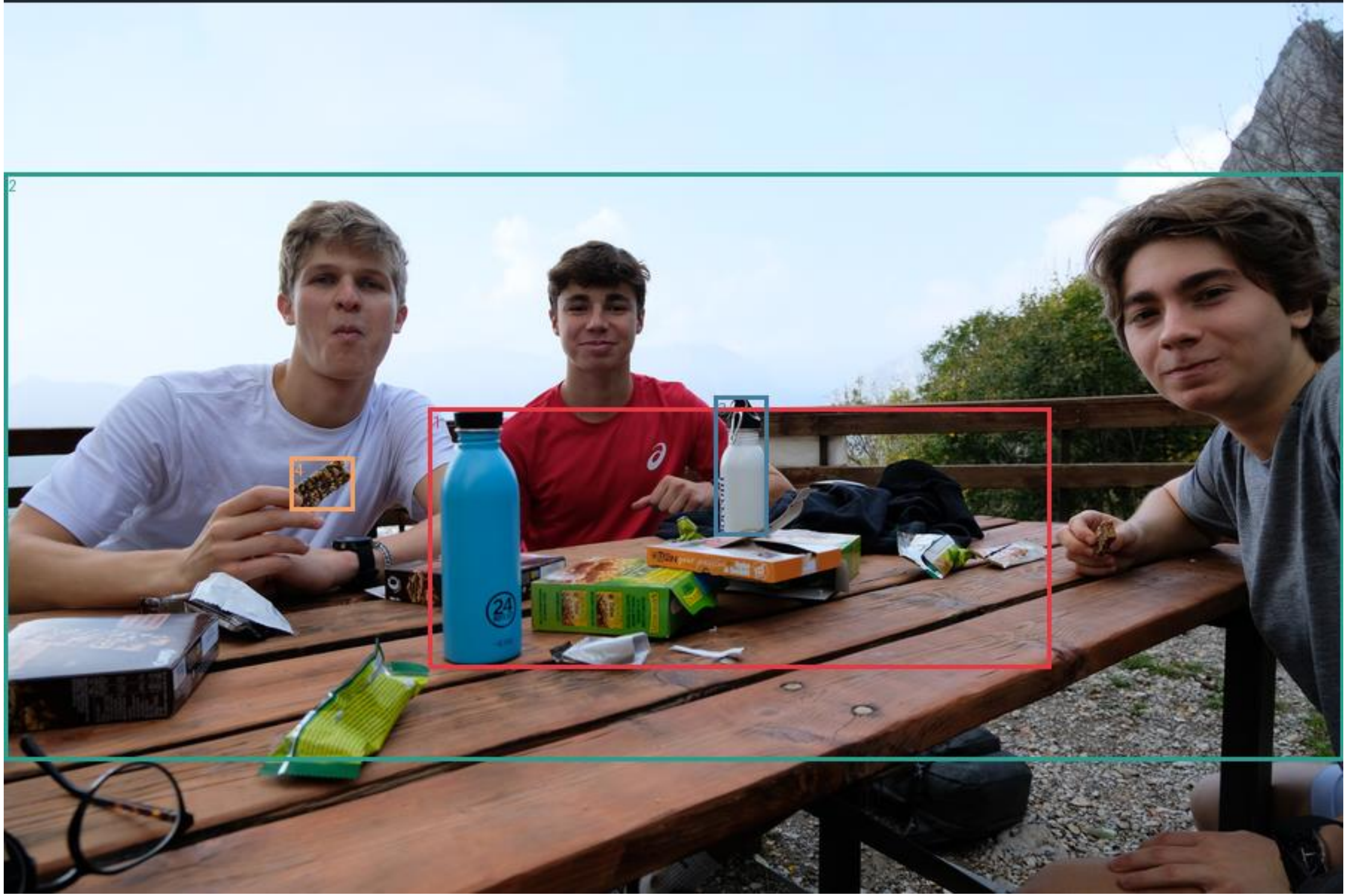


raw_crop_candidates | DSCF0136.JPG | 11 masks



crop_merged | DSCF0136.JPG | 2 masks

combined | DSCF0136.JPG | 4 masks



Apples-to-apples: same training, same eval

TRAINING PROTOCOL · Identical for both

Cascade Mask R-CNN

- Single GPU
- 20,000 iterations
- Batch size 8
- Learning rate 0.005

Only the pseudo-label set differs — baseline vs combined multi-scale.

EVALUATION · COCO val2017 (cls-agnostic)

The metrics that matter

APs — **AP at small-object scale**
primary indicator of small-object accuracy

AR-s — **Average Recall, small**
did the detector find them at all?

Optimizing AP vs AR is a different test — recall measures discovery, AP measures discovery + localization.

Three configurations compared: Pre-trained CutLER · Trained on baseline pseudo-labels · Trained on combined multi-scale pseudo-labels

FINAL RESULTS

Detector AP on COCO val2017 (BBOX)

Absolute AP is dominated by training-set scale (500 vs 1.3M images) — the level we care about is the small-object metric.

MODEL	AP	APs (small)	APm (medium)	APl (large)
Pre-trained CutLER ImageNet-1K (1.3M images)	12.33	3.66	13.45	29.60
Trained on baseline pseudo-labels TinyImageNet 500 images, single-scale MaskCut	2.22	1.40	2.72	4.00
Trained on combined pseudo-labels TinyImageNet 500 images, combined multi-scale MaskCut	3.07	1.20	4.19	4.87

KEY FINDING

Combined pseudo-labels improve overall AP significantly compared to baseline.
The multiscale masks helped segmentation of small objects but hurt bbox localization of small objects — likely because the tiny fragment masks (area ≤81px) introduced noisy bounding boxes that confused the bbox head.

THE HEADLINE · RECALL

~2× small-object recall

Combining multi-scale with baseline keeps medium and large coverage while also shifting the detector's attention toward small objects

AR small · THE WIN

0.040 → 0.084

baseline   combined multi-scale

how many of the true small objects in COCO were successfully recovered by the detector

from 4.0% to 8.4% of true small objects recovered

MODEL	AR small	AR medium	AR large
Baseline	**0.040**	0.114	0.020
Combined multi-scale	**0.084**	0.243	0.342
Multi-scale	0.078	0.013	0.000

Overall more small masks and more masks/image recovered.

HONEST ASSESSMENT

Limitations

- **Training set is 0.04% of the paper**

  500 images vs 1.3M — absolute AP is volume-bound, not algorithm-bound.

- **Multi-scale masks may be noisy**

  Median mask area is 45 px^2 — at this size, distinguishing object from texture is hard.
  (Maybe TinyImagenet and COCO is not ideal)

- **Time**

  On the 10-class TinyImageNet subset:
  - Single-scale MaskCut - **500 images in 58m, producing 790 annotations, ~7.0 seconds per image.**
  - Multi-scale - **7h30m, producing 3,094 multiscale annotations, ~54.4 seconds per image.**
  - **7.8×** slower while producing about **3.9×** more pseudo-masks.

Single-scale: 1 mask

Proposed small masks

CLOSING

Conclusions and Future Work

TAKEAWAY

Multi-scale MaskCut shifts the detector toward small objects

FUTURE WORK

01

Scale up training data

Move beyond 500 images toward the paper's 1.3M.

02

Get better masks

New/Improved way of proposing and filtering masks to deacrease the amount processing. Also better merging of overlappng masks.
Evaluate pseudo-label quality against COCO val2017 ground truth.

03

Different Dataset

Run training and evaluation with small object datasets

Thank you<3333